The Spanning Fan-connectivity and the Spanning Pipeline-connectivity of Graphs*

Cheng-Kuan Lin
Department of Computer Science
National Chiao Tung University, Hsinchu, Taiwan 30010, R.O.C.
cklin@cs.nctu.edu.tw

Jimmy J. M. Tan
Department of Computer Science
National Chiao Tung University, Hsinchu, Taiwan 30010, R.O.C.
jmtan@cs.nctu.edu.tw

D. Frank Hsu
Department of Computer and Information Science
Fordham University, New York, NY 10023, USA
hsu@trill.cis.fordham.edu

Lih-Hsing Hsu
Department of Computer Science and Information Engineering
Providence University, Taichung, Taiwan 43301, R.O.C.
lhhsu@pu.edu.tw

Abstract

Let G be a graph. The connectivity of G, $\kappa(G)$, is the maximum integer k such that there exists a k-container between any two different vertices. A k-container of G between u and v, $C_k(u,v)$, is a set of k-internally disjoint paths between u and v. A spanning container is a container that spans $V(G)$. A graph G is k^*-connected if there exists a spanning k-container between any two different vertices. The spanning connectivity of G, $\kappa^*(G)$, is the maximum integer k such that G is w^*-connected for $1 \leq w \leq k$ if G is 1^*-connected.

Let x be a vertex in G and let $U = \{y_1, y_2, \ldots, y_k\}$ be a subset of $V(G)$ where x is not in U. A spanning k-(x,U)-fan, $F_k(x,U)$, is a set of internally disjoint paths $\{P_1, P_2, \ldots, P_k\}$ such that P_i is a path connecting x to y_i for $1 \leq i \leq k$ and $\cup_{i=1}^{k} V(P_i) = V(G)$. A graph G is k^*-fan-connected (or k_f^*-connected) if there exists a spanning $F_k(x,U)$-fan for every choice of x and U with $|U| = k$ and $x \notin U$. The spanning fan-connectivity of a graph G, $\kappa_f^*(G)$, is defined as the largest integer k such that G is w_f^*-connected for $1 \leq w \leq k$ if G is 1_f^*-connected.

In this paper, some relationship between $\kappa(G)$, $\kappa^*(G)$, and $\kappa_f^*(G)$ are discussed. Moreover, some sufficient conditions for a graph to be k_f^*-connected are presented. Furthermore, we introduce the concept of a spanning pipeline-connectivity and discuss some sufficient conditions for a graph to be k^*-pipeline-connected.

*This work was supported in part by the National Science Council of the Republic of China under Contract NSC 94-2213-E-233-008.

1 Introduction

For graph definitions and notations, we follow [2]. Graph $G = (V, E)$ consists of a finite set V $(= (V(G))$ and a subset E $(= (E(G))$ of $\{(u,v) \mid u \neq v$ and (u,v) is an unordered pair of elements of $V\}$. We say that V is the *vertex set* and E is the *edge set*. We use $n(G)$ to denote $|V(G)|$. Two vertices u and v are *adjacent* if $(u,v) \in E$. A *vertex cut* is a set $S \subseteq V(G)$ such that $G-S$ has more than one component. A graph is *k-connected* if every vertex cut has at least k vertices. The *connectivity* of G, $\kappa(G)$, is the minimum size of a vertex cut. In other words, $\kappa(G)$ is the maximum k such that G is k-connected. A complete graph has no cut set. We adopt the convention that $\kappa(K_n) = n - 1$. A *path* is a sequence of vertices represented by $\langle v_0, v_1, \ldots, v_k \rangle$ with no repeated vertex and (v_i, v_{i+1}) is an edge of G for $0 \leq i \leq k-1$. We also write the path $\langle v_0, v_1, \ldots, v_k \rangle$ as $\langle v_0, \ldots, v_i, Q, v_j, \ldots, v_k \rangle$ where Q is a subpath from v_i to v_j. A *hamiltonian path* of a graph G is a path that contains all vertices of $V(G)$. A graph G is *hamiltonian connected* if there is a hamiltonian path between every two different vertices. A *cycle* is a path with at least three vertices such that the first vertex is the same as the last vertex. A *hamiltonian cycle* of G is a cycle that traverse every vertex of G. A graph is *hamiltonian* if it has a hamiltonian cycle. Let $P = \langle x_1, x_2, \ldots, x_k \rangle$ be a path of the graph G connecting x_1 and x_k. We use P^-1 to denote the path $\langle x_k, x_{k-1}, \ldots, x_1 \rangle$. We use $V(P)$ to denote the set $\{x_1, x_2, \ldots, x_k\}$ and $I(P)$ to denote the set $V(P) - \{x_1, x_k\}$. Let P_1 and P_2 be two paths of a graph G. We say that P_1 and P_2 are *internally-disjoint* if $I(P_1) \cap I(P_2) = \emptyset$.

Let u and v be two vertices of a graph G. A *k-container* of G between u and v, $C_k(u, v)$, is a set of k-internally disjoint paths between u and v [7]. It follows from Menger Theorem [13] that there is a k-container between any two distinct vertices of G if and only if G is k-connected. A k-container $C_k(u, v) = \{P_1, P_2, \ldots, P_k\}$ of G is a k^*-container if $\cup_{i=1}^{k} V(P_i) = V(G)$. A graph G is *k^*-connected* if there exists a k^*-container between any two distinct vertices. The *spanning connectivity* of G, $\kappa^*(G)$, is defined as the largest integer k such that G is w^*-connected for $1 \leq w \leq k$ if G is a 1^*-connected graph. It is obvious that a 1^*-connected graph is actually a hamiltonian connected graph and that a 2^*-connected graph is actually a hamiltonian graph. Moreover, any 1^*-connected graph except K_1 and K_2 is 2^*-connected. Thus, the concept of a k^*-connected graph is a hybrid concept of connectivity and hamiltonicity. Recently, the spanning connectivity of graphs have been studied extensively [1, 8, 9, 10, 11, 12, 16].

There is a Menger type theorem similar to the spanning connectivity of a graph. Let x be a vertex in a graph G and let $U = \{y_1, y_2, \ldots, y_t\}$ be a subset of $V(G)$ where x is not in U. A *t-(x, U)-fan*, $F_t(x, U)$, is a set of internally disjoint paths $\{P_1, P_2, \ldots, P_t\}$ such that P_i is a path connecting x and y_i for $1 \leq i \leq t$. It is proved by Dirac [6] that a graph G is k-connected if and only if it has at least $k+1$ vertices and there exists a t-(x, U)-fan for every choice of x and U with $|U| \leq k$ and $x \notin U$. Similarly, we can introduce the concept of a spanning fan. A *spanning k-(x, U)-fan* is a k-(x, U)-fan $\{P_1, P_2, \ldots, P_k\}$ such that $\cup_{i=1}^{k} V(P_i) = V(G)$. A graph G is *k^*-fan-connected* (also written as *k^*_f-connected*) if there exists a spanning k-(x, U)-fan for every choice of x and U with $|U| = k$ and $x \notin U$. The *spanning fan-connectivity* of a graph G, $\kappa^*_f(G)$, is defined as the largest integer k such that G is w^*_f-connected for $1 \leq w \leq k$ if G is a 1^*_f-connected graph. In this paper, some relationship among $\kappa(G)$, $\kappa^*(G)$, and $\kappa^*_f(G)$ are discussed. Moreover, some sufficient conditions for a graph to be k^*_f-connected are presented.

There is another Menger type theorem similar to the spanning connectivity and spanning fan-connectivity of a graph. Let $U = \{x_1, x_2, \ldots x_t\}$ and $W = \{y_1, y_2, \ldots, y_t\}$ be two t-subsets of $V(G)$. An *(U, W)-pipeline* is a set of internally disjoint paths $\{P_1, P_2, \ldots, P_t\}$ such that P_i is a path connecting x_i to $y_{\pi(i)}$ where π is a permutation of $\{1, 2, \ldots, t\}$. It is known that a graph G is k-connected if and only if it has at least $k+1$ vertices and there exists an (U, W)-pipeline for every choice of U and W with $|U| = |W| \leq k$ and $U \neq W$. Similarly, we can introduce the concept of spanning pipeline. A *spanning (U, W)-pipeline* is an (U, W)-pipeline $\{P_1, P_2, \ldots, P_k\}$ such that $\cup_{i=1}^{k} V(P_i) = V(G)$. A graph G is *k^*-pipeline-connected* (or *k^*_p-connected*) if there exists a spanning (U, W)-pipeline for every choice of U and W with $|U| = |W| \leq k$ and $U \neq W$. The *spanning pipeline-connectivity* of a graph G, $\kappa^*_p(G)$, is defined as the largest integer k such that G is w^*_p-connected for $1 \leq w \leq k$ if G is a 1^*_p-connected graph.

In Section 2, we establish some relationships among $\kappa(G)$, $\kappa^*(G)$, and $\kappa^*_f(G)$. Section 3 gives sufficient conditions for a graph to be k^*_f-connected. In Section 4, spanning pipeline-connectivity is included. Section 5 gives an example to illustrate the differences between $\kappa(G)$, $\kappa^*(G)$, $\kappa^*_f(G)$, and $\kappa^*_p(G)$.

2 Relationship among $\kappa(G)$, $\kappa^*(G)$, and $\kappa^*_f(G)$

A graph H is a *subgraph* of G if $V(H) \subseteq V(G)$ and $E(H) \subseteq E(G)$. Let S be a subset of G. The subgraph of G *induced* by S, denoted by $G[S]$, is the graph with the vertex set S and the edge set $\{(u, v) \mid (u, v) \in E(G) \text{ and } u, v \in S\}$. Let u be a vertex of G and let H be a subgraph of G. The *neighborhood* of u with respect to H, denoted by

$N_H(u)$, is $\{v \in V(H) \mid (u,v) \in E(G)\}$. We use $d_H(u)$ to denote $|N_H(u)|$. For any vertex u, the *degree* of u in G is $d_G(u)$. The *minimum degree* of G, written $\delta(G)$, is $\min\{d_G(x) \mid x \in V\}$. Let u and v be any two non-adjacent vertices of G, we use $G + (u,v)$ to denote the graph obtained from G by adding the edge (u,v).

Lemma 1. *Every 1^*-connected graph is 1^*_f-connected. Moreover, every 1^*_f-connected graph that is not K_2 is 2^*_f-connected. Thus, $\kappa^*_f(G) \geq 2$ if G is a hamiltonian connected graph with at least three vertices.*

Proof. Let G be a 1^*-connected graph with at least three vertices and let x be any vertex of G. Assume that $U = \{y\}$ with $x \neq y$. Obviously, there exists a hamiltonian path P_1 joining x and y. Apparently, $\{P_1\}$ forms a spanning 1-(x,U)-fan. Thus, G is 1^*_f-connected. Assume that $U = \{y_1, y_2\}$ with $x \notin U$. Let Q be a hamiltonian path of G connecting y_1 and y_2. We write Q as $\langle y_1, Q_1, x, Q_2, y_2\rangle$. We set P_1 as $\langle x, Q_1^{-1}, y_1\rangle$ and P_2 as $\langle x, Q_2, y_2\rangle$. Then $\{P_1, P_2\}$ forms a spanning 2-(x,U)-fan. Thus, G is 2^*_f-connected and $\kappa^*_f(G) \geq 2$. □

Theorem 1. *$\kappa^*_f(G) \leq \kappa^*(G) \leq \kappa(G)$ for any 1^*_f-connected graph. Moreover, $\kappa^*_f(G) = \kappa^*(G) = \kappa(G) = n(G) - 1$ if and only if G is a complete graph.*

Proof. Obviously, $\kappa^*(G) \leq \kappa(G)$. Now, we prove that $\kappa^*_f(G) \leq \kappa^*(G)$. Assume that $\kappa^*_f(G) = k$. Let x and y be any two vertices of G. We need to show that there is a k^*-container of G between x and y.

Suppose that $k = 1$. Since G is 1^*_f-connected, there is a spanning 1-$(x,\{y\})$-fan, $\{P_1\}$, of G. Then $\{P_1\}$ forms a spanning container of G between x and y.

Suppose that $k \geq 2$. Let $U' = \{y_1, y_2, \ldots, y_{k-1}\}$ be a set of $(k-1)$ neighbors of y not containing x. We set $U = U' \cup \{y\}$. By assumption, there exists a spanning k-(x,U)-fan. Obviously, we can extend the spanning k-(x,U)-fan by adding the edges $\{(y_i, y) \mid y_i \in U'\}$ to obtain a k^*-container between x to y. Hence, G is k^*-connected. Therefore, $\kappa^*_f(G) \leq \kappa^*(G)$ for every 1^*_f-connected graph.

Suppose that G is not a complete graph. There exists a vertex cut S of size $\kappa(G)$. Let x and y be any two vertices in different connected component of $G - S$. Obviously, y is not in any (x,S)-fan of G. Thus, $\kappa^*_f(G) < \kappa(G)$ □

3 Some sufficient conditions for a graph to be k^*_f-connected

Since the concept of spanning fan-connectivity is a generalization of hamiltonianicity, we review some previews results concerning hamiltonian graphs and hamiltonian connected graphs.

Lemma 2. *[5] Every graph G with at least three vertices and $\delta(G) \geq \frac{n(G)}{2}$ is 2^*-connected. Moreover, every graph G with at least four vertices and $\delta(G) \geq \frac{n(G)}{2} + 1$ is 1^*-connected.*

Lemma 3. *[14, 15] Let u and v be two non-adjacent vertices of G with $d_G(u) + d_G(v) \geq n(G)$. Then G is 2^*-connected if and only if $G + (u,v)$ is 2^*-connected. Moreover, suppose that $d_G(u) + d_G(v) \geq n(G) + 1$, then G is 1^*-connected if and only if $G + (u,v)$ is 1^*-connected.*

Lemma 4. *[3] A graph G is 2^*-connected if $d_G(u) + d_G(v) \geq n(G)$ for all non-adjacent vertices u and v. Moreover, a graph G is 1^*-connected if $d_G(u) + d_G(v) \geq n(G) + 1$ for all non-adjacent vertices u and v.*

For comparison reason, we list previews results concerning spanning connectivity.

Lemma 5. *[11] $\kappa^*(G) \geq 2\delta(G) - n(G) + 2$ if $\frac{n(G)}{2} + 1 \leq \delta(G) \leq n(G) - 2$.*

Lemma 6. *[12] Let k be a positive integer. Suppose that u and v are two nonadjacent vertices of G with $d_G(u) + d_G(v) \geq n(G) + k$. Then $\kappa^*(G) \geq k+2$ if and only if $\kappa^*(G + (u,v)) \geq k+2$.*

Lemma 7. *[12] Let k be a positive integer. Then $\kappa^*(G) \geq k + 2$ if $d_G(u) + d_G(v) \geq n(G) + k$ for all non-adjacent vertices u and v.*

Note that Lemma 5 (Lemma 6, and Lemma 7, respectively) generalizes the result of Lemma 2, (Lemma 3, and Lemma 4, respectively) in spanning connectivity. Result in the following theorem on spanning fan-connectivity is analogous to that on spanning connectivity in Lemma 6 [12].

Theorem 2. *Assume that k is a positive integer. Let u and v be two non-adjacent vertices of G with $d_G(u) + d_G(v) \geq n(G) + k$. Then $\kappa^*_f(G) \geq k + 1$ if and only if $\kappa^*_f(G + (u,v)) \geq k + 1$.*

Proof. Obviously, $\kappa^*_f(G + (u,v)) \geq k + 1$ if $\kappa^*_f(G) \geq k+1$. Suppose that $\kappa^*_f(G+(u,v)) \geq k+1$. Let x be any vertex of G and $U = \{y_1, y_2, \ldots, y_t\}$ be any subset of $V(G)$ such that $x \notin U$ and $t \leq$

$k+1$. We need to find a spanning t-(x,U)-fan of G.

Since $G+(u,v)$ is $(k+1)^*_f$-connected, there exists a spanning t-(x,U)-fan $\{P_1,P_2,\dots,P_t\}$ of $G+(u,v)$ with P_i joining x to y_i for $1\le i\le t$. Obviously, $\{P_1,P_2,\dots,P_t\}$ is a spanning t-(x,U)-fan of G if (u,v) is not in $\cup_{i=1}^t E(P_i)$. Thus, we consider (u,v) is in $\cup_{i=1}^t E(P_i)$. With Lemma 3, we can find a spanning (x,U)-fan of G if $t=1,2$. Thus, we consider the case $t\ge 3$. Without loss of generality, we may assume that $(u,v)\in P_1$. Thus, we can write P_1 as $\langle x,H_1,u,v,H_2,y_1\rangle$. Let P_i' be the path obtained from P_i by deleting x. Thus, we can write P_i as $\langle x,w_i,P_i',y_i\rangle$ for $1\le i\le t$. Note that $x\ne w_i$ and $P_i=\langle y_i\rangle$ if $w_i=y_i$ for every $2\le i\le t$.

Case 1: $d_{P_i'}(u)+d_{P_i'}(v)\ge n(P_i')+2$ for some $2\le i\le t$. Without loss of generality, we may assume that $d_{P_2'}(u)+d_{P_2'}(v)\ge n(P_2')+2$. Obviously, $n(P_2')\ge 2$. We write $P_2'=\langle w_2=z_1,z_2,\dots,z_r=y_2\rangle$. We claim that there exists an index j in $\{1,2,\dots,r-1\}$ such that $(z_j,v)\in E(G)$ and $(z_{j+1},u)\in E(G)$. Suppose that this is not the case. Then $d_{P_2'}(u)+d_{P_2'}(v)\le r+r-(r-1)=r+1=n(P_2')+1$. We get a contradiction.

We set $Q_1=\langle x,w_2=z_1,z_2,\dots,z_j,v,H_2,y_1\rangle$, $Q_2=\langle x,H_1,u,z_{j+1},z_{j+2},\dots,z_r=y_2\rangle$, and $Q_i=P_i$ for $3\le i\le t$. Then $\{Q_1,Q_2,\dots,Q_t\}$ forms a spanning t-(x,U)-fan of G. See Figure 1 for illustration.

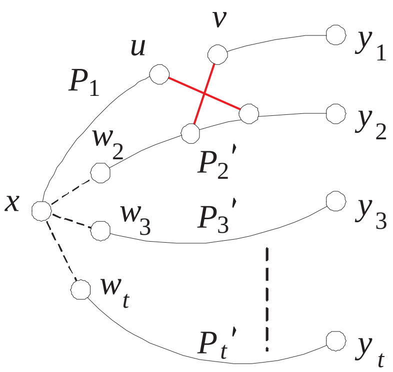

Figure 1: Illustration for case 1

Case 2: $d_{P_i'}(u)+d_{P_i'}(v)\le n(P_i')+1$ for every $2\le i\le t$.

Case 2.1: $d_{P_i'}(u)+d_{P_i'}(v)<n(P_i')+1$ for some $2\le i\le t$. Without loss of generality, we may assume that $d_{P_2'}(u)+d_{P_2'}(v)\le n(P_2')$. Thus,

$$
\begin{aligned}
& d_{P_1}(u)+d_{P_1}(v)\\
=\ & d_G(u)+d_G(v)-\sum_{i=2}^{t}(d_{P_i'}(u)+d_{P_i'}(v))\\
=\ & d_G(u)+d_G(v)-(d_{P_2'}(u)+d_{P_2'}(v))-\sum_{i=3}^{t}(d_{P_i'}(u)+d_{P_i'}(v))\\
\ge\ & n(G)+k-n(P_2')+\sum_{i=3}^{t}(n(P_i')+1)\\
=\ & n(P_1)+k-(t-2)\\
\ge\ & n(P_1)+1.
\end{aligned}
$$

With Lemma 3, there is a hamiltonian path Q_1 of $G[P_1]$ joining x to y_1. We set $Q_i=P_i$ for $2\le i\le t$. Then $\{Q_1,Q_2,\dots,Q_t\}$ forms a spanning t-(x,U)-fan of G. See Figure 2 for illustration.

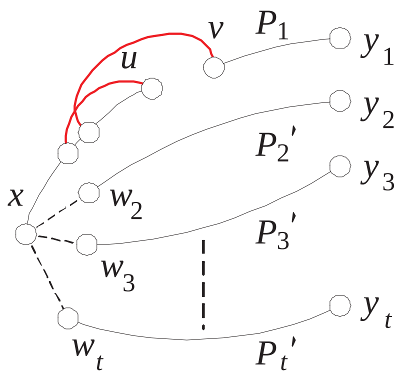

Figure 2: Illustration for case 2.1

Case 2.2: $d_{P_i'}(u)+d_{P_i'}(v)=n(P_i')+1$ for every $2\le i\le t$. We have

$$
\begin{aligned}
& d_{P_1}(u)+d_{P_1}(v)\\
=\ & d_G(u)+d_G(v)-\sum_{i=2}^{t}(d_{P_i'}(u)+d_{P_i'}(v))\\
=\ & n(G)+k-\sum_{i=2}^{t}(n(P_i')+1)\\
=\ & n(P_1)+k-(t-1)\\
\ge\ & n(P_1).
\end{aligned}
$$

We set $R=\langle y_1,H_2^{-1},v,u,H_1^{-1},x,w_2,P_2',y_2\rangle$. Then

$$
\begin{aligned}
& d_R(u)+d_R(v)\\
=\ & d_{P_1}(u)+d_{P_1}(v)+d_{P_2'}(u)+d_{P_2'}(v)\\
\ge\ & n(P_1)+n(P_2')+1\\
=\ & n(R)+1.
\end{aligned}
$$

With Lemma 3, there is a hamiltonian path W of $G[R]$ joining y_1 to y_2. Thus, W can be written as $\langle y_1, W_1, x, W_2, y_2\rangle$. We set $Q_1 = \langle x, W_1^{-1}, y_1\rangle$, $Q_2 = \langle x, W_2, y_2\rangle$, and $Q_i = P_i$ for $3 \le i \le t$. Then $\{Q_1, Q_2, \ldots, Q_t\}$ forms a spanning (x, U)-fan of G. □

We note that the Theorem 2 analogizes the result of Lemma 3 in spanning fan-connectivity.

Theorem 3. *Let k be a positive integer. Then $\kappa_f^*(G) \ge k+1$ if $d_G(u)+d_G(v) \ge n(G)+k$ for all non-adjacent vertices u and v.*

Note that Theorem 3 is an analog result of Lemma 4 in spanning fan-connectivity.

Theorem 4. *$\kappa_f^*(G) \ge 2\delta(G) - n(G) + 1$ if $\frac{n(G)}{2} + 1 \le \delta(G)$.*

Proof. Suppose that $\frac{n(G)}{2} + 1 \le \delta(G)$. Obviously, $\delta(G) \le n(G) - 1$ and $n(G) \ge 4$. Suppose that $n(G) = 2m$ for some integer $m \ge 2$. Then $\delta(G) = m+k$ for some integer k with $1 \le k \le m-1$. Obviously, $d_G(u) + d_G(v) \ge 2\delta(G) = 2m + 2k$. By Theorem 3, $\kappa_f^*(G) \ge 2k+1 = 2\delta(G)-n(G)+1$. Suppose that $n(G) = 2m + 1$ for some integer $m \ge 2$. Then $\delta(G) = m + 1 + k$ for some integer k with $1 \le k \le m-1$. By Theorem 3, $\kappa_f^*(G) \ge 2k+2 = 2\delta(G) - n(G) + 1$. The theorem is proved. □

The Theorem 4 analogizes the result of Lemma 2 in spanning fan-connectivity. Moreover, when $\delta(G) = n(G) - 2$, we have the following corollary.

Corollary 1. *$\kappa_{fan}^*(G) = n(G) - 3$ if $\delta(G) = n(G) - 2$ and $n(G) \ge 5$.*

Proof. By Lemma 5, $\kappa^*(G) \ge n(G) - 2$. Since $n(G) - 2 \le \kappa^*(G) \le \kappa(G) \le \delta(G) = n(G) - 2$, $\kappa(G) = n(G) - 2$. By Theorem 4, $\kappa_{fan}^*(G) \ge n(G) - 3$. By Theorem 1, $\kappa_{fan}^*(G) < \kappa(G)$. Thus, $\kappa_{fan}^*(G) = n(G) - 3$. □

4 Spanning pipeline-connectivity

Similar to some recent works on the spanning connectivity [1, 8, 9, 10, 11, 12, 16] and the spanning fan-connectivity, studied in Section 2, we study spanning pipeline-connectivity in this section. Lemma 8 and Theorem 5 are analogous to Lemma 1 and Theorem 1 respectively.

Lemma 8. *Every 1^*-connected graph is 1_p^*-connected.*

Theorem 5. *$\kappa_p^*(G) \le \kappa_f^*(G) \le \kappa^*(G) \le \kappa(G)$ for any 1_p^*-connected graph. Moreover, $\kappa_p^*(G) = \kappa_f^*(G) = \kappa^*(G) = \kappa(G)$ if and only if G is a complete graph.*

Theorem 6. *Assume that k is a positive integer. Let u and v be two non-adjacent vertices of G. Suppose that $d_G(u) + d_G(v) \ge n(G) + k$. Then $\kappa_p^*(G) \ge k$ if and only if $\kappa_p^*(G + (u, v)) \ge k$.*

Proof. Obviously, $\kappa_p^*(G + (u, v)) \ge k$ if $\kappa_p^*(G) \ge k$. Suppose that $\kappa_p^*(G + (u, v)) \ge k$. Let $U = \{x_1, x_2, \ldots, x_t\}$ and $W = \{y_1, y_2, \ldots, y_t\}$ be any two subsets of G such that $U \ne W$ and $t \le k$. We need to find a spanning (U, W)-pipeline of G.

Since $G + (u, v)$ is k_p^*-connected, there exists a spanning (U, W)-pipeline of $G + (u, v)$. Let $\{P_1, P_2, \ldots, P_t\}$ be a spanning (U, W)-pipeline with P_i joining x_i to $y_{\pi(i)}$ for $1 \le i \le t$. Without loss of generality, we assume that $\pi(i) = i$. Obviously, $\{P_1, P_2, \ldots, P_t\}$ is a spanning (U, W)-pipeline of G if (u, v) is not in P. Thus, we consider the case that (u, v) is in P. By Lemma 3, we can find a spanning (U, W)-pipeline of G if $t = 1$. Thus, we consider the case $t \ge 2$.

Case 1: $U \cap W = \emptyset$. Without loss of generality, we may assume that $(u, v) \in P_1$. Thus, we can write P_1 as $\langle x_1, H_1, u, v, H_2, y_1\rangle$. (Note that $H_1 = \langle x\rangle$ if $x = u$, and $H_2 = \langle y\rangle$ if $y = v$.) Let P_i' be the path obtained from P_i by deleting x and y_i. Thus, we can write P_i as $\langle x_i, P_i', y_i\rangle$ for $1 \le i \le t$.

Case 1.1: $d_{P_1}(u) + d_{P_1}(v) \ge n(P_1) + 1$. With Lemma 3, there is a hamiltonian path Q_1 of $G[P_1]$ joining x_1 to y_1. We set $Q_i = P_i$ for $2 \le i \le t$. Then $\{Q_1, Q_2, \ldots, Q_t\}$ forms a spanning (U, W)-pipeline of G. See Figure 3 for illustration.

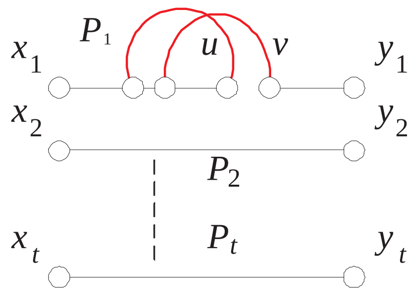

Figure 3: Illustration for Case 1.1.

Case 1.2: $d_{P_1}(u)+d_{P_1}(v) \le n(P_1)$. We claim that $d_{P_i}(u) + d_{P_i}(v) \ge n(P_i) + 2$ for some $2 \le i \le t$.

Suppose that $d_{P_i}(u) + d_{P_i}(v) \le n(P_i) + 1$ for every $2 \le i \le t$. Then

$$\begin{aligned} & d_G(u) + d_G(v) \\ = \;& d_{P_1}(u) + d_{P_1}(v) + \sum_{i=2}^{t} (d_{P_i}(u) + d_{P_i}(v)) \end{aligned}$$

$$\begin{aligned} &\leq n(P_1) + \sum_{i=2}^{t}(n(P_i)+1) \\ &= n(G) + t - 1 \\ &\leq n(G) + k - 1. \end{aligned}$$

We obtain a contradiction. Thus, $d_{P_i}(u) + d_{P_i}(v) \geq n(P_i) + 2$ for some $2 \leq i \leq t$. Without loss of generality, we assume that $d_{P_2}(u) + d_{P_2}(v) \geq n(P_2) + 2$. Obviously, $n(P_2') \geq 2$. We write $P_2' = \langle x_2 = z_1, z_2, \ldots, z_r = y_2\rangle$. We claim that there exists an index j in $\{1, 2, \ldots, r-1\}$ such that $(z_j, v) \in E(G)$ and $(z_{j+1}, u) \in E(G)$. Suppose this is not the case. Then $d_{P_2'}(u) + d_{P_2'}(v) \leq r + r - (r-1) = r + 1 = n(P_2') + 1$. We get a contradiction.

We set $Q_1 = \langle x_2 = z_1, z_2, \ldots, z_j, v, H_2, y_1\rangle$, $Q_2 = \langle x_1, H_1, u, z_{j+1}, z_{j+2}, \ldots, z_r = y_2\rangle$, and $Q_i = P_i$ for $3 \leq i \leq t$. Then $\{Q_1, Q_2, \ldots, Q_t\}$ forms a spanning (U, W)-pipeline of G. See Figure 4 for illustration.

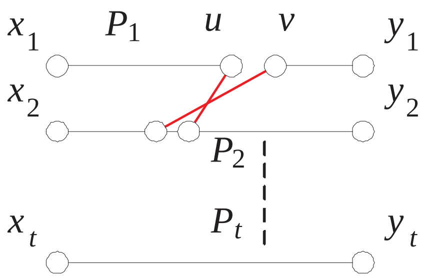

Figure 4: Illustration for Case 1.2.

Case 2: $U \cap W \neq \emptyset$. Let $|U \cap W| = r$. Without loss of generality, we assume that $x_i = y_i$ for $t - r + 1 \leq i \leq t$. Let $G' = G[V(G) - (U \cap W)]$, $U' = U - W$, and $W' = W - U$. Obviously, $d_{G'}(u) + d_{G'}(v) \geq d_G(u) + d_G(v) - 2r \geq n(G) + k - 2r = n(G') + k - r$, $|U'| = |W'| = t - r \leq k - r$, and $U' \cap W' = \emptyset$. By Case 1, there exists a spanning (U', W')-pipeline $\{Q_1, Q_2, \ldots, Q_{t-r}\}$ of G'. We set $Q_i = \langle x_i\rangle$ for $t - r + 1 \leq i \leq t$. Then $\{Q_1, Q_2, \ldots, Q_t\}$ forms a spanning (U, W)-pipeline of G. $\square$

We note that Theorem 6, Theorem 7, and 8 are analogous to Lemma 3, Lemma 4, and Lemma 2 in spanning pipeline-connectivity respectively.

Theorem 7. *Let k be a positive integer. Then $\kappa_p^*(G) \geq k$ if $d_G(u) + d_G(v) \geq n(G) + k$ for all non-adjacent vertices u and v.*

Theorem 8. $\kappa_p^*(G) \geq 2\delta(G) - n(G)$ *if* $\frac{n(G)}{2} + 1 \leq \delta(G)$.

Corollary 2. $\kappa_p^*(G) = n(G) - 4$ *if* $\delta(G) = n(G) - 2$ *and* $n(G) \geq 5$.

Proof. By Theorem 8, $\kappa_p^*(G) \geq n(G) - 4$. Let $V(G) = \{x_1, x_2, \ldots, x_{n(G)}\}$. Without loss of generality, we assume that $(x_1, x_2) \notin E(G)$. We set $U = \{x_3, x_5, x_6, \ldots, x_{n(G)}\}$ and $W = \{x_4, x_5, x_6 \ldots, x_{n(G)}\}$. Obviously, $U \neq W$ and $|U| = |W| = n(G) - 3$. It is easy to check that $G[V(G) - \{x_5, x_6, \ldots, x_{n(G)}\}]$ is isomorphic to a cycle with four vertices. Since there is not a hamiltonian path of $G[V(G) - \{x_5, x_6, \ldots, x_{n(G)}\}]$ joining x_1 to x_3, there is not a spanning (U, W)-pipeline of G. Thus, $\kappa_p^*(G) = n(G) - 4$. $\square$

5 An example

We use the following example to illustrate that $\kappa(G)$, $\kappa^*(G)$, $\kappa_f^*(G)$, and $\kappa_p^*(G)$ are really different concepts and, in general, have different values .

Example 1. *Suppose that n is a positive integer with $n \geq 2$. Let $H(n)$ be the complete 3-partite graph $K_{n,n,n-1}$ with vertex partite sets $V_1 = \{x_1, x_2, \ldots x_{2n}\}$, $V_2 = \{y_1, y_2, \ldots, y_{2n}\}$, and $V_3 = \{z_1, z_2, \ldots, z_{n-1}\}$. Let $G(n)$ be the graph obtained from $H(n)$ by adding the edge set $\{(z_i, z_j) \mid 1 \leq i \neq j < n\}$. Thus, $G[V_3]$ is the complete graph K_{n-1}. Obviously, $n(G(n)) = 5n - 1$, $\delta(G(n)) = 2n + (n-1) = 3n - 1$, and $\kappa(G(n)) = \delta(G(n))$. In the following, we will show that $\kappa^*(G(n)) = n + 1$, $\kappa_f^*(G(n)) = n$, and $\kappa_p^*(G(n)) = n - 1$.*

By Lemma 5, $\kappa^*(G(n)) \geq 2\delta(G(n)) - n(G(n)) + 2 = n + 1$. To show $\kappa^*(G(n)) = n + 1$, we claim that there is no $(n+2)^*$-container of $G(n)$ between x_1 and x_2. Suppose this is not the case. Let $\{P_1, P_2, \ldots, P_{n+2}\}$ be an $(n+2)^*$-container of $G(n)$ between x_1 and x_2. Obviously, $|V(P_i) \cap (V_1 - \{x_1, x_2\})| \leq |V(P_i) \cap (V_2 \cup V_3)| - 1$ for $1 \leq i \leq n+2$. Thus $\sum_{i=1}^{n+2} |V(P_i) \cap (V_1 - \{x, y\})| = (\sum_{i=1}^{n+2} |(V_2 \cup V_3) \cap V(P_i)|) - (n+2)$. Therefore, $|V_1 - \{x, y\}| \leq |V_2 \cup V_3| - (n+2)$. However, $|V_1 - \{x, y\}| = 2n - 2$ but $|V_2 \cup V_3| - (n+2) = 2n - 3$. This leads to a contradiction.

By Theorem 4, $\kappa_f^*(G(n)) \geq 2\delta(G(n)) - n(G(n)) + 1 = n$. To show $\kappa_f^*(G(n)) = n$, we claim that there is no spanning (x_1, U)-fan of $G(n)$ where $U = \{y_1, y_2, \ldots, y_{n+1}\}$. Suppose not. Let $\{P_1, P_2, \ldots, P_{n+1}\}$ be a spanning (x_{n+1}, U)-fan of $G(n)$. Without loss of generality, we assume that P_i is a path joining x to y_i for $1 \leq i \leq n+1$. Obviously, $|(V_1 - \{x\}) \cap V(P_i)| \leq |((V_2 \cup V_3) - \{y_i\}) \cap V(P_i)|$. Thus, $\sum_{i=1}^{n+1} |(V_1 - \{x\}) \cap V(P_i)| \leq$

$\sum_{i=1}^{n+1} |((V_2 \cup V_3) - \{y_i\}) \cap V(P_i)|$. *Therefore,* $|V_1 - \{x\}| \leq |(V_2 \cup V_3) - U|$. *However,* $|V_1 - \{x\}| = 2n - 1$ *but* $|(V_2 \cup V_3) - U| = 2n - 2$. *We get a contradiction.*

By Theorem 8, $\kappa_p^*(G(n)) \geq 2\delta(G(n)) - n(G(n)) = n - 1$. *To prove* $\kappa_p^*(G(n)) = n - 1$, *we claim that there is no spanning* (U, W)*-pipeline where* $U = \{x_1, x_2, \ldots, x_n\}$ *and* $W = \{x_{n+1}, x_{n+2}, \ldots, x_{2n}\}$. *Suppose that there exists a spanning* (U, W)*-pipeline* $\{P_1, P_2, \ldots, P_n\}$. *Obviously,* $|V_2 \cap V(P_i)| - 1 \leq |V_3 \cap V(P_i)|$ *for* $1 \leq i \leq n$. *Then* $\sum_{i=1}^{n}(|V_2 \cap V(P_i)| - 1) \leq \sum_{i=1}^{n} |V_3 \cap V(P_i)|$. *Therefore,* $(\sum_{i=1}^{n} |V_2 \cap V(P_i)|) - n \leq \sum_{i=1}^{n} |V_3 \cap V(P_i)|$. *However,* $(\sum_{i=1}^{n} |V_2 \cap V(P_i)|) - n = |V_2| - n = n$ *but* $\sum_{i=1}^{n} |V_3 \cap V(P_i)| = |V_3| = n - 1$. *We get a contradiction.*

References

[1] M. Albert, R.E.L. Aldred, and D. Holton, On 3*-connected graphs, Australasian Journal of Combinatorics 24 (2001) 193–208.

[2] J.A. Bondy and U.S.R. Murty, Graph Theory with Applications, North Holland, New York, 1980.

[3] J.A. Bondy and V.Chvátal, A method in graph theory, Discrete Mathematic 15 (1976) 111-135 .

[4] C.-H. Chang, C.-K. Lin, H.-M. Huang, and L.-H. Hsu, The super laceability of the hypercubes, Information Processing Letters 92 (2004) 15–21.

[5] G.A. Dirac, Some theorem on abstract graphs, Proceedings of the London Mathematical Society 2 (1952) 69–81.

[6] G.A. Dirac, In abstrakten Graphen vorhandene vollstäandige 4-Graphen und ihre Unterteilungen, Mathematische Nachrichten 22 (1960) 61–85.

[7] D. F. Hsu, On container width and length in graphs, groups, and networks, IEICE Transactions Fundamentals E77-A (1994) 668–680.

[8] H.-C. Hsu, C.-K. Lin, H.-M. Hung, and L.-H. Hsu, The spanning connectivity of the (n,k)-star graphs, International Journal of Foundations of Computer Science 17 (2006) 415–434.

[9] C.-K. Lin, H.-M. Huang, and L.-H. Hsu, The super connectivity of the pancake graphs and star graphs, Theoretical Computer Science 339 (2005) 257–271.

[10] C.-K. Lin, H.-M. Huang, D.F. Hsu, and L.-H. Hsu, The spanning diameter of the star graph, Networks 48 (2006) 235–249.

[11] C.-K. Lin, H. M.-Huang, and L.-H. Hsu, On the spanning connectivity of graphs, Discrete Mathematics 307 (2007) 285–289.

[12] C.-K. Lin, H.-M. Huang, J.J.M. Tan, and L.-H. Hsu, On spanning connected graphs, Discrete Mathematics *to appear.*

[13] K. Menger, Zur allgemeinen kurventheorie, Fundamentale Mathematik 10 (1927) 95–115.

[14] O. Ore, Note on Hamilton circuits, American Mathematical Monthly 67 (1960) 55.

[15] O. Ore, Hamiltonian connected graphs, Journal of Mathematic Pures Application 42 (1963) 21-27.

[16] C.-H. Tsai, J.J.M. Tan, and L.-H. Hsu, The super connected property of recursive circulant graphs, Information Processing Letters 91 (2004) 293–298.